

SUNDERMERE HEALTH DATA CONSORTIUM. STATISTICAL DISCLOSURE STANDARD, EDITION 4 (SDS-4). Extract for member bodies. Fictional consortium, fictional standard, fictional registry, invented for testing. The numbers in this standard are invented for this exercise and are not anybody's real published threshold. The rules and the wording below are the authoritative rulebook for this exercise. Treat them as correct and complete, and do not rely on outside knowledge of disclosure control, of any published cell size, or of any other body's rules.

SECTION 1. WHICH RULEBOOK GOVERNS, AND WHY THAT HAS TO BE SETTLED FIRST.

1.1 There is no universal minimum cell size. Bodies that publish health, education and research statistics each set their own, and the figures they publish run from single digits to the thirties. At least one major statistical body's stated position is that no univocal criterion exists at all. A number carried over from another body's rulebook is not a rule here. A cell size means nothing until the rulebook it belongs to has been named, and naming that rulebook is the first act of any review, before a single value is looked at.

1.2 This standard, SDS-4, governs every table a member body publishes from data it holds under its Consortium data use agreement. Compliance is a term of that agreement. It is an obligation, not a guideline.

1.3 Where a table is built from data supplied by another body under that body's own agreement, that body's rules govern the table and this standard does not. Wrensfeld holds Vellacourt Tumour Registry Partnership data under a separate agreement, and the Vellacourt release rules govern any table built from it. Those rules are reproduced at section 8 and their minimum cell size is not the same as this standard's. What the table was built from decides, not who is publishing it and not what the table is about.

SECTION 2. MINIMUM CELL SIZE.

2.1 The minimum cell size under this standard is 11. A cell whose value is 1 to 10 inclusive is a small cell and must not be published. 11 is the first reportable value: a cell of exactly 11 satisfies this standard and is published. A cell of 11 is not a borderline case and is not rounded, moved or withheld out of caution.

2.2 A value of zero does not violate the minimum cell size, and a zero cell is published. A zero says that no individual in the group has the attribute. It identifies nobody, because there is nobody in it to be identified. Suppressing a zero costs information and buys no protection, and this standard does not require it, whatever the zero happens to say about the body publishing it.

2.3 The rule runs on the count of individuals in the cell, not on how the cell is displayed. See section 4.

SECTION 3. SUPPRESSION, AND WHY ONE SUPPRESSION IS USUALLY NOT ENOUGH.

3.1 A small cell is suppressed. That suppression is the primary suppression.

3.2 A primary suppression protects nothing if the suppressed value can be recovered. A suppressed value must not be derivable by addition or subtraction from the other cells published in the same table, taken together with any marginal total published alongside them. Where it is derivable, one or more further cells must be suppressed as well. Those are complementary suppressions.

3.3 The commonest failure under this standard is a breakdown into two categories with a total. Suppressing the small category leaves the other category and the total both published, and the total minus the other category is the suppressed value, exactly, to the individual. The suppression has achieved nothing at all. It has told the reader which cell was sensitive and handed them its value in the same breath. Either the second category is suppressed as a complementary, or the marginal total is not published.

3.4 Complementary suppressions may not be needed if marginal totals are not published. Where the total is withheld there is nothing to subtract from, and a single primary suppression can be sufficient on its own. Withholding the margin is a permitted alternative to a complementary suppression, and which of the two costs less information is a judgement for the reviewer.

3.5 A zero cell is useless as a complementary suppression. The union of a sensitive cell and a zero cell is the sensitive cell: the two suppressed together sum to the sensitive value, so subtracting the published cells from the total returns that value unchanged. A complementary suppression must be a cell with a value in it. Choosing the zero because suppressing a zero appears to cost nothing is the error this rule exists to name.

3.6 Aggregating adjacent categories, so that the small cell no longer exists as a category, is a permitted alternative to suppression, and where it is available it is usually preferred, because it withholds nothing. A category combined with its neighbour must produce a cell that satisfies section 2 on its own.

SECTION 4. PERCENTAGES, RATES AND ANY OTHER FORMULA.

4.1 A percentage published with its base is a count. The count is the percentage multiplied by the base, and it is recoverable to the individual. Section 2 applies to that count exactly as it applies to a cell of counts. Republishing a small cell as a percentage does not take it out of this standard, and a small denominator is where this bites hardest, because a percentage over a small base carries almost no rounding to hide behind.

4.2 Every percentage, rate, ratio or index published under this standard must be published with the base it was computed over. A share without a denominator is not a statistic. It follows that the base cannot be withheld in order to rescue a percentage that fails section 2. That cell has to be dealt with under section 3.

4.3 No published figure, formula or derived value may display a suppressed value or allow it to be reconstructed. This covers a total, a difference, an average multiplied by a count, a rate, and a percentage over a published base.

SECTION 5. RECODING AND TOP CODING.

5.1 Recoding the extreme of a distribution, by merging the top or the bottom categories into a wider one, is a permitted way to remove a small cell, subject to 3.6.

5.2 Recoding one end of a distribution is not enough on its own where the remaining categories and the base are still published. The reported categories subtracted from a hundred hand the recoded category's share straight back, and the base turns that share into a count. A recode at the high end that leaves every other category and the base published has protected nobody. One end is never enough.

5.3 It follows that a recode is checked against 3.2 like any other suppression, and that a recoded category which is still a small cell is still a small cell. Merging two small categories into one category that is also small has changed the label and nothing else.

SECTION 6. A RELEASE IS NOT ASSESSED ON ITS OWN.

6.1 A table is assessed against every table the body has already published from the same population, and not only against itself.

6.2 Two releases that each satisfy section 2 on their own can together isolate a cell that satisfies neither. Where the same population is reported twice under different groupings, the difference between the two groupings is a cell that neither release published and both releases disclose. The commonest form by far is a change to the band boundaries between one release and the next, made because the bands read better, by somebody who never compared the two.

6.3 A table that isolates a small cell against a prior release cannot be published as drafted, whatever it looks like on its own. The remedy is to keep the prior release's boundaries, or to combine categories under 3.6 so that no new boundary is created, or to suppress under section 3. The prior release is public and cannot be recalled, so the draft is the only one of the two that can move.

SECTION 7. MAGNITUDE DATA.

7.1 A magnitude cell holds a value contributed by identifiable units rather than a count of individuals. Payments to businesses, amounts billed and amounts written off are magnitude data.

7.2 A magnitude cell must satisfy both of the following.

(a) The number of contributors to the cell is at least the minimum cell size at 2.1.

(b) The two largest contributors to the cell together account for 60 per cent or less of the cell's value.

7.3 The test at 7.2(b) is the dominance rule and it is the one that matters. A cell can have forty contributors, far past 7.2(a), and still be disclosive: where a small number of them contribute most of the value, the published total is a usable estimate of the largest contributor, and the count of contributors has protected nobody. A large contributor count is never a substitute for the dominance test. The dominance test is computed from the contributor values. It is not judged by eye and it is not inferred from how many contributors there are.

7.4 Where a magnitude cell fails 7.2 it is suppressed, and section 3 applies to it exactly as it applies to a cell of counts.

SECTION 8. THE VELLACOURT RELEASE RULES, REPRODUCED.

8.1 These are the Vellacourt Tumour Registry Partnership's own release rules, reproduced here for member bodies that hold Vellacourt data. They govern every table built from Vellacourt data, in place of sections 2 to 7 of this standard wherever the two differ.

8.2 The minimum cell size under the Vellacourt rules is 16. A cell whose value is 1 to 15 inclusive must not be published. 16 is the first reportable value.

8.3 A value of zero does not violate the Vellacourt minimum, and a zero cell is published.

8.4 The Vellacourt rules adopt the suppression, complementary suppression, percentage, recoding, prior release and magnitude rules at sections 3 to 7 of this standard without change. Only the minimum cell size is different.

8.5 A table built from Vellacourt data is not made compliant by satisfying SDS-4. The two minimums are different numbers, and that difference is the entire reason this section exists.